

A Cybersecurity Incident Reporting And Risk Assessment System For Detecting, And Prioritizing Security Threats In An Organization

A CAPSTONE PROJECT REPORT

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IN

ELECTRONICS AND COMMUNICATION ENGINEERING

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September -2026



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DECLARATION

We M. Janaki (192412409) N. Sanjana (192412478) Syed Nazma (192425193) students of Bachelor of Engineering in electronics and communication and Engineering, Saveetha University of Medical and Technical Sciences, hereby declare that the work presented in this capstone project titled: **A Cybersecurity Incident Reporting and Risk Assessment System for Detecting, and Prioritizing Security Threats in an Organization** is the outcome of our own knowledge and effort. This work has been carried out with due consideration of engineering ethics.

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**A Cybersecurity Incident Reporting and Risk Assessment System for Detecting, and Prioritizing Security Threats in an Organization**” has been carried out by M. Janaki (192412409 N. Sanjana (192412478) Syed Nazma (192425193) under the supervision of **Dr. S.K.Saravanan** is submitted in partial fulfilment of the requirements for the current semester of the B.Tech ECE program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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ABSTRACT

Cybersecurity incidents have become a major concern for organizations because security threats such as phishing, malware, unauthorized access, password attacks, data breaches, and suspicious network activities can affect the confidentiality, integrity, and availability of organizational information. When security incidents are not reported and evaluated properly, it becomes difficult for an organization to determine which threats require immediate attention. Therefore, an organized incident reporting and risk assessment system is required to record security incidents, evaluate their potential impact, and prioritize threats according to their level of risk.

This project proposes a **Cybersecurity Incident Reporting and Risk Assessment System for Detecting and Prioritizing Security Threats in an Organization**. The proposed system provides a structured method for recording cybersecurity incidents and assessing their risk using parameters such as incident type, likelihood, impact, and severity. The system categorizes reported incidents into different risk levels such as Low, Medium, High, and Critical so that security personnel can identify important threats and respond to them in an appropriate order. The system can also maintain incident information using file handling so that previously reported incidents can be stored and reviewed when required.

The implementation is designed using the **C programming language**, making use of structures, functions, conditional statements, arrays, file handling, and menu-driven programming. The system consists of three major modules: incident reporting, risk assessment, and threat prioritization and reporting. The risk assessment module calculates a risk score based on the likelihood and impact of an incident, while the prioritization module arranges incidents according to their risk level. The proposed system provides a simple and practical approach for demonstrating cybersecurity risk management concepts in an organizational environment.

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CHAPTER 1

INTRODUCTION

1.1 Background Information

Organizations depend heavily on computers, networks, databases, cloud services, and digital communication systems for their daily activities. As the amount of digital information increases, organizations also face different types of cybersecurity threats. Phishing attacks, malware infections, unauthorized access, weak passwords, data theft, and suspicious activities can cause financial loss, service interruption, and loss of confidential information.

An important part of cybersecurity management is the ability to report and manage incidents in an organized manner. When an employee or system identifies a security incident, important information such as the incident type, affected system, date, description, and severity should be recorded. The recorded information can then be analysed to determine the possible risk to the organization.

Risk assessment helps organizations understand the likelihood and potential impact of security threats. NIST Cybersecurity Framework 2.0 provides guidance for organizations to understand, assess, prioritize, and communicate cybersecurity risks. It also includes risk-assessment outcomes related to identifying threats, vulnerabilities, likelihood, impact, and risk responses.

The proposed project applies these basic principles to develop a simple cybersecurity incident reporting and risk assessment system. The system is implemented using C programming and is designed to demonstrate how security incidents can be recorded, evaluated, classified, and prioritized.

Figure 1.1: General cybersecurity incident management process

Insert a diagram showing:

Security Incident



Incident Reporting



Incident Information



Risk Assessment



Risk Score



Threat Classification



Prioritization



Security Action

1.2 Project Objectives

The main objective of the project is to develop a simple cybersecurity incident reporting and risk assessment system that helps an organization record security incidents and prioritize threats based on their risk.

The specific objectives are to develop a structured incident reporting mechanism, store incident information using file handling, evaluate the likelihood and impact of reported incidents, calculate a risk score, classify incidents according to their severity, and prioritize high-risk incidents for immediate attention.

Another objective is to demonstrate how basic cybersecurity risk management concepts can be implemented using C programming techniques such as structures, functions, conditional statements, arrays, and file handling.

1.3 Significance

The proposed system is significant because it provides a simple method for organizing cybersecurity incident information. Instead of maintaining incident information manually, the system allows users to enter and store incident details in a structured format.

Risk prioritization is also important because organizations may receive multiple security incidents at the same time. Treating every incident with the same priority may result in important threats being overlooked. By assigning risk levels, the proposed system helps users identify which incidents should receive greater attention.

The approach also supports the basic principles of modern cybersecurity risk management. NIST CSF 2.0 is intended to help organizations manage cybersecurity risk regardless of their size or level of cybersecurity maturity.

1.4 Scope

The project focuses on developing a software-based cybersecurity incident reporting and risk assessment system. The system allows users to enter information about security incidents and evaluate their risk based on predefined criteria.

The system covers incident recording, risk-score calculation, risk classification, incident storage, and threat prioritization. It is designed as an academic prototype and does not attempt to replace professional Security Information and Event Management (SIEM), Security Operations Center (SOC), or enterprise incident-response platforms.

The project can be further expanded in the future by adding network monitoring, automated threat detection, database storage, user authentication, email alerts, and graphical dashboards.

1.5 Methodology Overview

The project methodology begins with identifying common cybersecurity incidents and determining the information required to report them. A suitable incident-record structure is then designed to store details such as incident ID, incident type, description, likelihood, impact, risk score, and risk level.

The risk assessment process uses the likelihood and impact values to calculate a risk score. Based on the resulting score, the incident is classified into an appropriate risk category. The incident records are stored using C file-handling techniques so that they can be accessed later.

Finally, the system displays the reported incidents and prioritizes them according to their risk level. Testing is performed using different incident scenarios to verify whether the system produces the expected classification and prioritization.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Organizations can experience several cybersecurity incidents involving employees, computers, networks, applications, and data. If these incidents are reported without a standard format, important information may be missing. This makes it difficult for security personnel to understand the seriousness of an incident.

Another problem is that not all security incidents have the same level of risk. For example, a minor suspicious email and a confirmed data breach should not necessarily receive the same priority. Therefore, organizations require a systematic method to evaluate the likelihood and impact of incidents.

The proposed system addresses this problem by providing a structured reporting process followed by risk assessment and threat prioritization.

2.2 Evidence of the Problem

Cybersecurity incident response is an important part of organizational cybersecurity risk management. NIST's current SP 800-61 Rev. 3 explains that incident response should be incorporated throughout cybersecurity risk-management activities and that effective incident response can help reduce the number and impact of incidents.

The NIST Cybersecurity Framework 2.0 also identifies the importance of recording threats and evaluating potential likelihood and impact so that inherent risk can be understood and used for prioritizing risk responses.

These principles demonstrate the need for organizations to maintain proper incident information and evaluate security risks rather than simply recording incidents without further analysis.

2.3 Stakeholders

The main stakeholders of the proposed system include employees, IT administrators, cybersecurity personnel, system administrators, and organizational management.

Employees can use the system to report suspicious activities or security incidents. IT and cybersecurity personnel can review the reported incidents and determine their risk levels. Management can use summarized information to understand the most important security problems and support appropriate security decisions.

The system can also be useful for students and academic institutions as a practical demonstration of cybersecurity incident management and risk assessment.

2.4 Supporting Data/Research

Modern cybersecurity guidance emphasizes the importance of identifying, assessing, prioritizing, and communicating cybersecurity risks. NIST CSF 2.0 provides a framework for managing cybersecurity risk, while the 2025 revision of NIST SP 800-61 integrates incident response recommendations with the CSF 2.0 risk-management approach.

NIST CSF 2.0 specifically includes risk-assessment outcomes related to identifying threats and vulnerabilities, determining likelihood and impact, understanding inherent risk, and prioritizing risk responses.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

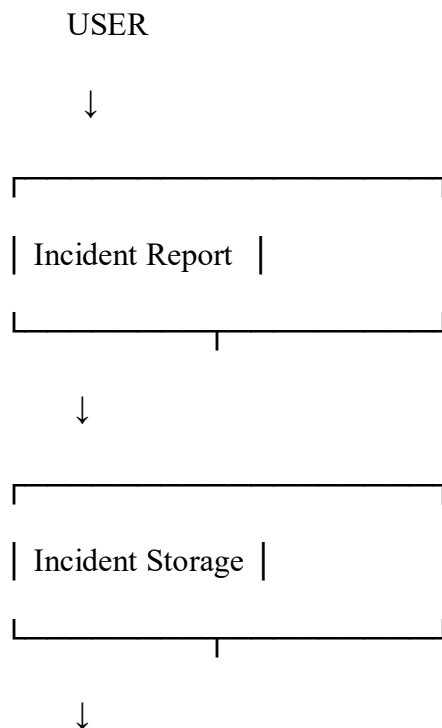
3.1 Development & Design Process

The development process begins by identifying the information that needs to be collected when a cybersecurity incident occurs. An incident record is designed using a C structure containing fields such as incident ID, incident type, description, likelihood, impact, risk score, and risk level.

The system is then divided into separate functional modules. The incident reporting module accepts information from the user. The risk assessment module evaluates the likelihood and impact of the incident and calculates its risk score. The prioritization module classifies and arranges incidents according to their risk level.

File handling is used to store incident records so that information is not lost when the program is closed. Functions are used to keep the program organized and make each operation easier to test and maintain.

Figure 3.1: Proposed system architecture





3.2 Tools & Technologies Used

The primary programming language used for the implementation is **C**. C provides the basic programming features required to create the system, including structures, functions, arrays, conditional statements, loops, and file handling.

A standard C compiler or IDE can be used for development and testing. File handling functions such as `fopen()`, `fprintf()`, `fscanf()`, `fread()`, `fwrite()`, and `fclose()` can be used depending on the selected storage method.

The main technologies and concepts used are:

- C programming
- Structures

- Functions
- Arrays
- Conditional statements
- Loops
- File handling
- Risk scoring
- Threat classification
- Menu-driven programming

3.3 Solution Overview

The proposed system contains three main modules.

Module 1 – Cybersecurity Incident Reporting

The user enters information about the detected security incident. The information can include the incident ID, incident type, affected system, description, date, likelihood, and impact.

Examples of incident types include:

- Phishing
- Malware
- Unauthorized access
- Password attack
- Data leakage
- Suspicious activity

The information is stored as an incident record.

Module 2 – Risk Assessment

The system evaluates the reported incident based on **likelihood** and **impact**.

A simple risk-score calculation can be used:

Risk Score = Likelihood × Impact

For example, if:

Likelihood = 4

and

Impact = 5

then:

Risk Score = $4 \times 5 = 20$

The score can then be converted into a risk category according to the defined thresholds.

Module 3 – Threat Prioritization and Reporting

After risk scores are calculated, incidents are categorized into levels such as:

Risk Score Risk Level

1–4	Low
5–9	Medium
10–15	High
16–25	Critical

The exact thresholds can be adjusted during implementation, but they must remain consistent throughout the system.

Critical incidents receive the highest priority, followed by High, Medium, and Low incidents.

Figure 3.2: Risk assessment process

Incident

↓

Likelihood

+

Impact

↓

Risk Score

↓

1–4 → LOW	
5–9 → MEDIUM	
10–15 → HIGH	
16–25 → CRITICAL	

↓

Priority

3.4 Engineering Standards Applied

The project follows basic software engineering practices such as modular design, structured programming, meaningful variable names, proper input validation, and systematic testing.

The cybersecurity risk-assessment concept is aligned with the principles of NIST Cybersecurity Framework 2.0. The framework provides a common approach for understanding, assessing, prioritizing, and communicating cybersecurity risks.

Incident-management concepts are also aligned at a high level with NIST SP 800-61 Rev. 3, which integrates incident response with broader cybersecurity risk management.

3.5 Ethical Standards Applied

Cybersecurity systems must be designed and used responsibly. The proposed system is intended for authorized incident reporting and risk assessment within an organization.

The system should not be used to attack systems, obtain unauthorized information, or access confidential data. Test incidents should be simulated or generated using authorized information.

The project also follows ethical principles of accurate reporting. Incident information should not be deliberately modified to produce a particular risk level, and system results should be documented honestly.

3.6 Solution Justification

The proposed solution is suitable because it provides a simple and structured method for handling cybersecurity incidents. It combines incident reporting, risk assessment, and threat prioritization into a single system.

The use of C programming makes the project suitable for demonstrating structured programming and file-handling concepts while applying them to a cybersecurity problem.

The approach is also consistent with current cybersecurity risk-management principles. NIST CSF 2.0 emphasizes the need to understand and prioritize cybersecurity risks, while NIST SP 800-61 Rev. 3 emphasizes integrating incident response into cybersecurity risk management.

CHAPTER 4

RESULT & RECOMMENDATIONS

4.1 Evaluation of Results

The system can be evaluated by entering different types of cybersecurity incidents and assigning different likelihood and impact values.

For example, a low-impact incident with low likelihood should produce a low risk score, while an incident with high likelihood and high impact should receive a critical risk classification.

The system should correctly store incident information, calculate the risk score, assign the appropriate risk category, and display incidents according to their priority.

Figure 4.1: Sample risk assessment output

Insert your actual C-program output screenshot here.

Figure 4.2: Sample threat-prioritization output

Insert your actual program output showing Low, Medium, High, and Critical incidents here.

The final report should use the **actual output generated by your program** rather than invented results.

4.2 Challenges Encountered

One challenge during development is designing a suitable structure for storing all the required incident information. Another challenge is handling incorrect user input and ensuring that the program does not produce invalid risk scores.

File handling can also create difficulties because records must be written and retrieved correctly. Another challenge is ensuring that the risk classification thresholds remain consistent for all reported incidents.

These problems can be addressed through modular programming, input validation, repeated testing, and checking each function separately.

4.3 Possible Improvements

The proposed system can be improved by adding a database instead of simple file storage. A graphical user interface can also be developed to make the system easier for non-technical users.

Future versions can include user authentication, automated email notifications, real-time network monitoring, automated log analysis, and dashboard-based visualization.

Machine learning techniques could also be introduced in a future version to identify patterns in historical incident records and assist with automated threat classification.

4.4 Recommendations

Organizations should encourage employees to report suspicious activities as soon as they are detected. Incident information should be recorded using a consistent format so that security personnel can analyse incidents effectively.

Risk assessment should be performed regularly because the likelihood and impact of threats can change over time. High and critical incidents should receive priority for investigation and response.

The proposed prototype should also be extended with stronger authentication and secure storage before being considered for use with real organizational security information.

CHAPTER- 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

5.1.1 Academic Knowledge

The project provided a better understanding of important cybersecurity concepts such as security threats, vulnerabilities, cybersecurity incidents, risk assessment, incident response, and threat prioritization. Before developing the system, cybersecurity incidents were mainly understood as individual security problems. Through this project, it became clear that an organization also needs a systematic process for reporting, analysing, and managing these incidents. The project helped in understanding how likelihood and impact can be used to evaluate the seriousness of a security threat and how the resulting risk information can support decision-making.

The project also provided knowledge about cybersecurity risk-management frameworks and their importance in organizations. NIST Cybersecurity Framework 2.0 provides guidance for organizations to assess, prioritize, and communicate cybersecurity risks. This helped in understanding how the proposed system can be related to real-world cybersecurity practices.

5.1.2 Technical Skills

The project improved practical programming skills using the **C programming language**. Different C programming concepts such as structures, functions, arrays, conditional statements, loops, and file handling were applied while developing the system. Structures were useful for representing incident records, while functions helped divide the program into smaller and manageable modules.

File handling provided practical experience in storing and retrieving incident information. Input validation was also considered to reduce incorrect entries. Developing the project as separate modules helped improve understanding of how a larger programming problem can be divided into smaller tasks and combined into a complete working system.

5.1.3 Problem-Solving and Critical Thinking

The project required several decisions regarding how cybersecurity incidents should be represented and evaluated. One important problem was determining which information should be collected when an incident is reported. Another was deciding how likelihood and impact could be combined to produce a meaningful risk score.

Testing different combinations of likelihood and impact helped in understanding how a small change in an input value can change the final risk classification. The project also required thinking about how multiple incidents could be compared and prioritized. This improved analytical thinking and helped develop a more systematic approach to solving technical problems.

5.2 Challenges Encountered and Overcome

Several challenges were encountered during the development of the system. One of the initial challenges was designing an appropriate structure for storing incident information. The system needed to maintain different details such as incident type, description, likelihood, impact, risk score, and risk level in an organized manner.

Another challenge was implementing file handling correctly so that incident records could be saved and retrieved when required. Input validation was also important because incorrect values could produce incorrect risk assessments. These challenges were addressed by dividing the program into smaller functions and testing each function individually before integrating the complete system.

The risk-classification rules also required careful testing. Different sample incidents were entered into the system to verify whether the calculated risk score resulted in the expected Low, Medium, High, or Critical category.

5.3 Applications of Engineering Standards

The project demonstrated the importance of applying structured engineering practices during software development. Modular programming was used to separate incident reporting, risk assessment, storage, and prioritization functions. Proper input validation and systematic testing were also considered to improve the reliability of the system.

The cybersecurity risk-assessment approach was developed with reference to **NIST Cybersecurity Framework 2.0**, which provides a common approach for organizations to understand, assess, prioritize, and communicate cybersecurity risks. The framework also identifies the importance of recording threats, determining likelihood and impact, understanding inherent risk, and using this information to prioritize risk responses.

Therefore, the project helped demonstrate how engineering principles and cybersecurity standards can be incorporated into a simple software-based solution.

5.4 Applications of Ethical Standards

Ethical considerations are important when working with cybersecurity information because incident reports may contain sensitive organizational details. The proposed system is intended only for authorized incident reporting and risk assessment. It should not be used to access systems without permission, obtain confidential information, or perform malicious activities.

The project also highlighted the importance of maintaining accurate incident information. Security personnel should report incidents honestly and should not modify information simply to reduce or increase the calculated risk level. Confidential information should also be protected from unauthorized access.

Through this project, the importance of responsible cybersecurity practices and ethical use of technical knowledge was understood more clearly.

5.5 Insights into the Industry

The project provided a better understanding of how cybersecurity is handled in real organizational environments. Cybersecurity is not limited to preventing attacks. Organizations must also be able to identify incidents, record relevant information, assess their possible impact, prioritize them, and coordinate an appropriate response.

Current NIST guidance emphasizes integrating incident response with broader cybersecurity risk management. NIST SP 800-61 Revision 3 explains that incident-response activities should be incorporated throughout cybersecurity risk-management activities to improve incident detection, response, and recovery.

The project also showed the importance of prioritization. When an organization has several incidents, security teams cannot necessarily treat every incident with the same urgency. Incidents need to be categorized and prioritized according to their potential risk. This is directly reflected in NIST CSF 2.0, which includes incident triage, categorization, and prioritization as part of incident management.

This provided useful insight into the type of processes that are followed in cybersecurity-related roles.

5.6 Conclusion of Personal Development

Overall, the project contributed to both technical and personal development. It improved knowledge of cybersecurity concepts while also providing practical experience in C programming, file handling, modular programming, testing, and system design.

The project also improved problem-solving and critical-thinking abilities because different technical problems had to be analysed and solved during development. Preparing the report and documenting the system further improved technical communication skills.

Most importantly, the project helped connect academic concepts with a practical cybersecurity application. The experience gained from developing the incident reporting, risk assessment, and threat prioritization modules can provide a useful foundation for further learning in cybersecurity, software development, network security, and information technology.

CHAPTER 6

CONCLUSION

6.1 Summary of Key Findings

The project developed a **Cybersecurity Incident Reporting and Risk Assessment System** for recording, evaluating, and prioritizing security incidents within an organization.

The system uses three main functions: incident reporting, risk assessment, and threat prioritization. Incident information is recorded and stored, while likelihood and impact are used to calculate a risk score. Based on the score, incidents are categorized into Low, Medium, High, or Critical levels.

The implementation demonstrates that basic C programming concepts such as structures, functions, conditional statements, and file handling can be applied to develop a practical cybersecurity management prototype.

6.2 Impacts and Significance

The proposed system provides a simple approach for organizing cybersecurity incidents and identifying threats that require greater attention. It can help demonstrate the importance of systematic incident reporting and risk assessment in an organization.

The project is also aligned with current cybersecurity risk-management principles. NIST CSF 2.0 provides a framework for organizations to understand, assess, prioritize, and communicate cybersecurity risk, while SP 800-61 Rev. 3 provides updated guidance for integrating incident response with cybersecurity risk management.

Although the developed system is an academic prototype, its basic structure can be extended into a more advanced cybersecurity platform by integrating databases, authentication, automated detection, network monitoring, dashboards, and alert mechanisms.

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APPENDICES:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Admin Dashboard</title>
```

```
<style>
```

```
body {  
    margin: 0;  
    font-family: Arial, sans-serif;  
    background: #f4f6f8;  
}
```

```
/* HEADER */
```

```
.header {  
    background: #17202a;  
    color: white;  
    padding: 25px 40px;  
}
```

```
.header h1 {  
    margin: 0;  
}
```

```
.header p {  
    margin-bottom: 0;  
}
```

```
/* MAIN */
```

```
.container {  
  padding: 30px;  
}
```

```
/* ANALYTICS CARDS */
```

```
.cards {  
  display: flex;  
  gap: 20px;  
  margin-bottom: 30px;  
}
```

```
.card {  
  background: white;  
  flex: 1;  
  padding: 20px;  
  text-align: center;  
  border-radius: 10px;  
  box-shadow: 0 2px 8px #ccc;  
}
```

```
.card h2 {  
  font-size: 30px;  
  margin: 5px;  
}
```

```
.card p {  
  color: #666;  
}
```

```
/* ANALYSIS */
```

```
.analysis-box {  
    background: white;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 2px 8px #ccc;  
    margin-bottom: 30px;  
}
```

```
.analysis-box p {  
    font-size: 17px;  
    margin: 12px 0;  
}
```

```
/* FILTER BOX */
```

```
.filter-box {  
    background: white;  
    padding: 20px;  
    border-radius: 10px;  
    box-shadow: 0 2px 8px #ccc;  
    margin-bottom: 30px;  
}
```

```
input,  
select {  
    padding: 10px;  
    margin-right: 10px;  
    border: 1px solid #ccc;  
    border-radius: 5px;  
}
```

```
.filter-button {  
    padding: 10px 18px;  
    background: #2874a6;  
    color: white;  
    border: none;  
    border-radius: 5px;  
    cursor: pointer;  
}
```

```
/* TABLE */
```

```
.table-box {  
    background: white;  
    padding: 25px;  
    border-radius: 10px;  
    box-shadow: 0 2px 8px #ccc;  
    overflow-x: auto;  
}
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
}
```

```
th {  
    background: #17202a;  
    color: white;  
    padding: 13px;  
}
```

```
td {  
    padding: 12px;  
    text-align: center;
```

```
border-bottom: 1px solid #ddd;  
}
```

```
/* RISK COLORS */
```

```
.critical {  
  color: red;  
  font-weight: bold;  
}
```

```
.high {  
  color: orange;  
  font-weight: bold;  
}
```

```
.medium {  
  color: #b8860b;  
  font-weight: bold;  
}
```

```
.low {  
  color: green;  
  font-weight: bold;  
}
```

```
/* STATUS */
```

```
.status {  
  font-weight: bold;  
}
```



```
/* BUTTONS */
```

```
.view-button {  
  background: #2874a6;  
  color: white;  
  padding: 7px 12px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
.update-button {  
  background: #27ae60;  
  color: white;  
  padding: 7px 12px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
.logout {  
  margin-top: 25px;  
  background: #555;  
  color: white;  
  padding: 11px 20px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
a {  
  text-decoration: none;  
}
```

```
/* STATUS FORM */

.status-form {
    margin-top: 7px;
}

.status-form select {
    padding: 6px;
    margin: 0;
}

</style>

</head>

<body>

<!-- HEADER -->

<div class="header">

    <h1>🛡️ Cybersecurity Admin Dashboard</h1>

    <p>Incident Monitoring and Risk Assessment System</p>

</div>
```

```
<div class="container">
```

```
<!-- ANALYTICS CARDS -->
```

```
<div class="cards">
```

```
<div class="card">
```

```
<h2>{{ total }}</h2>
```

```
<p>Total Incidents</p>
```

```
</div>
```

```
<div class="card">
```

```
<h2>{{ critical }}</h2>
```

```
<p>Critical Incidents</p>
```

```
</div>
```

```
<div class="card">
```

```
<h2>{{ high }}</h2>
```

```
<p>High Risk Incidents</p>
```

```
</div>
```

<div class="card">

<h2>{{ resolved }}</h2>

<p>Resolved Incidents</p>

</div>

</div>

<!-- RISK ANALYSIS -->

<div class="analysis-box">

<h2> Risk Analysis</h2>

<p>

 Critical:

{{ critical }}

</p>

<p>

 High:

{{ high }}

</p>

<p>

 Medium:

{{ medium }}

</p>

<p>

□ Low:

{{ low }}

</p>

</div>

<!-- FILTER -->

<div class="filter-box">

<h2>🔍 Search and Filter</h2>

<form method="GET" action="/admin">

<input

type="text"

name="search"

placeholder="Search incident..."

value="{{ search }}"

>

<select name="risk">

<option value="">

All Risk Levels

</option>

```
<option value="CRITICAL"
  {% if risk == "CRITICAL" %}selected{% endif %}>
  Critical
</option>

<option value="HIGH"
  {% if risk == "HIGH" %}selected{% endif %}>
  High
</option>

<option value="MEDIUM"
  {% if risk == "MEDIUM" %}selected{% endif %}>
  Medium
</option>

<option value="LOW"
  {% if risk == "LOW" %}selected{% endif %}>
  Low
</option>

</select>
```

```
<select name="status">
```

```
  <option value="">
```

```
    All Status
```

```
  </option>
```

```
  <option value="Pending"
```

```
    {% if status == "Pending" %}selected{% endif %}>
```

```
    Pending
```

```
  </option>
```

```
<option value="Investigating"
  {% if status == "Investigating" %}selected{% endif %}>
  Investigating
</option>
```

```
<option value="Resolved"
  {% if status == "Resolved" %}selected{% endif %}>
  Resolved
</option>
```

```
</select>
```

```
<button
  type="submit"
  class="filter-button"
>
  APPLY FILTER
</button>
```

```
</form>
```

```
</div>
```

```
<!-- INCIDENT TABLE -->
```

```
<div class="table-box">
```

```
<h2>&img alt="lock icon" data-bbox="181 823 201 838"/> Reported Security Incidents</h2>
```

```
{% if incidents %}
```

```
<table>
```

```
<tr>
```

```
<th>ID</th>
```

```
<th>Title</th>
```

```
<th>Type</th>
```

```
<th>System</th>
```

```
<th>Likelihood</th>
```

```
<th>Impact</th>
```

```
<th>Risk Score</th>
```

```
<th>Risk Level</th>
```

```
<th>Status</th>
```

```
<th>Actions</th>
```

```
</tr>
```

```
{% for incident in incidents %}
```

```
<tr>
```

```
<!-- ID -->
```



```
<td>
    {{ incident[0] }}
</td>
```

```
<!-- TITLE -->
```

```
<td>
    {{ incident[1] }}
</td>
```

```
<!-- TYPE -->
```

```
<td>
    {{ incident[2] }}
</td>
```

```
<!-- SYSTEM -->
```

```
<td>
    {{ incident[4] }}
</td>
```

```
<!-- LIKELIHOOD -->
```

```
<td>
    {{ incident[5] }}
</td>
```

```
<!-- IMPACT -->
```

```
<td>
```

```
  {{ incident[6] }}
```

```
</td>
```

```
<!-- RISK SCORE -->
```

```
<td>
```

```
  <b>
```

```
    {{ incident[7] }}
```

```
  </b>
```

```
</td>
```

```
<!-- RISK LEVEL -->
```

```
<td>
```

```
{% if incident[8] == "CRITICAL" %}
```

```
  <span class="critical">
```

```
    CRITICAL
```

```
  </span>
```

```
{% elif incident[8] == "HIGH" %}
```

```
  <span class="high">
```

```
    HIGH
```

```
  </span>
```

```

{% elif incident[8] == "MEDIUM" %}

    <span class="medium">
        MEDIUM
    </span>

{% else %}

    <span class="low">
        LOW
    </span>

{% endif %}

</td>

<!-- STATUS -->

<td class="status">

    {{ incident[9] }}

</td>

<!-- ACTIONS -->

<td>

    <a
        href="/incident/{{ incident[0] }}"
    >

```

```

        <button class="view-button">
            VIEW
        </button>

    </a>

    <form
        method="POST"
        action="/update_status/{{ incident[0] }}"
        class="status-form"
    >

        <select name="status">

            <option value="Pending">
                Pending
            </option>

            <option value="Investigating">
                Investigating
            </option>

            <option value="Resolved">
                Resolved
            </option>

        </select>

        <button
            type="submit"
            class="update-button"
        >

```

UPDATE

</button>

</form>

</td>

</tr>

{% endfor %}

</table>

{% else %}

<h3>No incidents found.</h3>

<p>

Try changing the search or filter options.

</p>

{% endif %}

</div>

<!-- LOGOUT -->


```
<button class="logout">
```

```
    LOGOUT
```

```
</button>
```

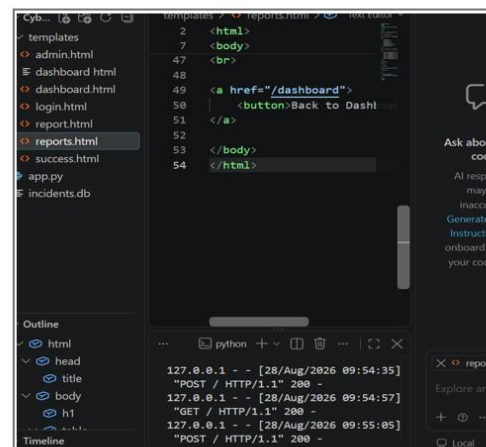
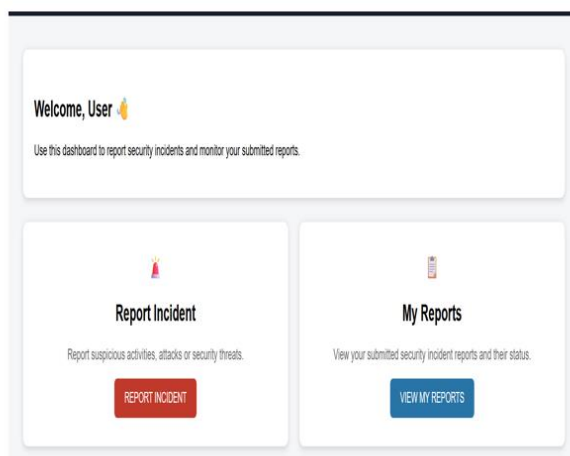
```
</a>
```

```
</div>
```

```
</body>
```

```
</html>
```

OUTPUT:



Incident Reporting OUTPUT:

Incident Information

Incident Title
Example: Phishing Attack

Incident Type
Select Incident Type

Description
Describe the security incident...
Please fill in this field.

Affected System
Example: Employee PC

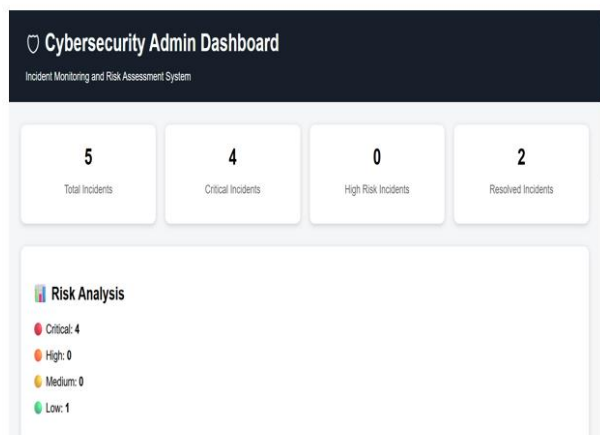
Likelihood (1 - 5)
1 - Very Low

Impact (1 - 5)
1 - Very Low

```
templates > success.html > Text Editor
2 <html>
7 <body>
18
19 <a href="/report">
20 | <button>Report Another
21 </a>
22
23 <br><br>
24
25 <a href="/">
26 | <button>Logout</button>
27 </a>
28
29 </body>
30 </html>
```

```
127.0.0.1 - - [28/Aug/2026 09:54:35]
"POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:54:57]
"GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:55:05]
```

Admin Dashboard Output:



```
admin.html > dashboard > Text Editor
222 <html>
223 <body>
237 <div class="container">
399 <div class="table-box">
598 |
599 | {% endif %}
600 </div>
601
602 </div>
603
604
605 <!-- LOGOUT -->
606
607 <a href="/">
608 | <button class="logout">
609 | | LOGOUT
610 | </button>
611 </a>
612
613
```

```
127.0.0.1 - - [28/Aug/2026 09:54:35]
"POST / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:54:57]
"GET / HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 09:55:05]
"POST / HTTP/1.1" 200 -
```